

Satvik Sharma

Punjab, India

+91 7340936506 | satviksharma511@gmail.com

[LinkedIn: Satvik Sharma](#) | [Github: Satvik-Sharma511](#)

Education

VIT Bhopal University

B.Tech Computer Science and Engineering (AIML), CGPA: 8.19/10.0

September 2023 – May 2027

Sehore, MP

Saint Peter Senior Secondary School

Class XII (10+2)

2023

Chandigarh

Saint Soldier's Divine Public School

Class X

2021

Panchkula, Haryana

Technical Skills

Programming & Web Technologies: C++, Python, JavaScript, HTML, CSS, React.js, RESTful APIs

Databases & Cloud platforms: SQL, MongoDB, AWS, Docker, Git, GitHub

AI/ML & MLOps: Scikit-learn, XGBoost, Pandas, Numpy, LangChain, RAG, DVC, MLflow, DagsHub

UI/UX Design & Productivity Tools: UI/UX Design, Figma, Microsoft Office (Excel, PowerPoint)

Experience

FOSSEE Semester Long Internship (IIT Bombay)

February 2026 – June 2026

Software Engineer Intern

Remote

- Delivered core full-stack features – spanning a React.js frontend and Node.js/Express.js backend – for a web-based, open-source electronics simulation and learning platform aiming to serve 1000+ concurrent users.
- Built role-based dashboard systems, classroom management workflows, and a gamified progression system, increasing weekly user engagement by 30% and improving platform scalability.

Projects

SolarCarbon AI | *Python, FastAPI, Mendeley Data* | [Deployed Link](#)

- Developed a platform that predicts users earned **carbon credits** from their renewable energy usage (Solar panels).
- Trained a Voting Regressor ensemble of **3 models** – HistGradientBoosting, RandomForest, and ExtraTrees – on a self-authored Mendeley dataset of **230+ thermal plant emission records** to predict grid emission factors through physics-based feature engineering.
- Constructed a preprocessing pipeline combining **2 techniques** – RobustScaler and PolynomialFeatures – to model non-linear relationships and achieve an R^2 score of **0.97**.

OpenHW Studio | *React.js, Node.js, Express.js, MongoDB, TypeScript, Docker* | [Deployed Link](#)

- Engineered a full-stack, browser-based electronics simulation platform under FOSSEE, IIT Bombay, contributing across **4 areas** – frontend, backend, the TypeScript emulator, and documentation – and assembling **twelve virtual hardware components** from scratch spanning the manifest, logic, and UI layers, plus fixes to telemetry.
- Architected the Arduino Adventure Map, a full-stack gamification system structured around **2 mechanics** – linear component unlocking and XP/leveling – and implemented an automated assessment engine combining topological circuit-wiring analysis with regex-based C++ static code analysis to autonomously grade submissions against an **80% threshold**, saving **15+ instructor hours** weekly.
- Designed **3 MongoDB schemas** – UserProgress, QuizAttempts, and CompletedProjects – secured JWT-authenticated REST endpoints, and established the platform's foundational UI/UX along with a **14-step**, accessibility-compliant Tour Guide onboarding system.

RAG-Powered AI Code Reviewer | *LangChain, ChromaDB, Sentence Transformers* | [Deployed Link](#)

- Deployed an automated code reviewer as a **GitHub Webhook** that solves the “institutional memory” gap in tools like GitHub Copilot, autonomously posting context-aware review summaries on new Pull Requests via **RAG**.
- Implemented a **FastAPI** backend to extract and chunk PR diffs via the GitHub API, retrieving relevant precedents from a **ChromaDB** vector store powered by Sentence Transformers (**all-MiniLM-L6-v2**).
- Orchestrated a **LangChain + Groq** (llama-3.1-8b) generation pipeline that posts JSON-structured feedback within **5 seconds** of a Pull Request opening.

Achievements

- Hackathon Finalist:** Advanced to the Buildathon Final Review round out of 100+ competing teams in NASSCOM and Gnani.ai's “Automation using Agentic AI” challenge, building an intelligent AI agent with Inya.ai for Consumer Durables.
- Carbon Emission Dataset:** Published on Mendeley Data, adding and verifying latitude-longitude columns across 1,000+ records to support carbon credit research.

Certifications

- Applied ML in Python – University of Michigan, Coursera (2025)
- NPTEL Cloud Computing – IIT Kharagpur (2025)
- JLPT N5 - Japanese Language Proficiency Test (2025)
- Figma UI/UX Design – Udemy (2024)
- Postman API – Postman (2024)